ACTL 2131/5101 - Assignment 2 (practical)

Submission deadline:

Friday, 24th April, 6 pm sharp via Turnitin on Moodle

This assessment activity aims to reinforce your learning by providing you with an opportunity to apply the concepts you have learned in the course to real life problems.

Instructions:

- This is an individual assignment.
- You should perform your analysis in R.
- Each assignment should be typed in.
- Assignments must be submitted via Turnitin on Moodle. The submission consists of two parts:
  - **Part 1: Report** (maximum 2 pages). Any content beyond page 2 will not be marked. Please be concise while clearly addressing each question.
  - **Part 2: R code** submitted as a single file, with clear comments indicating where each question begins.
- Please ensure that all work is your own, as plagiarism will be detected by Turnitin.

Background/Data preparation

You are an analyst working for an investment bank. Your manager has requested you to download, prepare and analyse data concerning financial returns. The data should contain at least 1,000 observations and should be **merged by date** across different websites/data files.

UPDATE: S&P 500 ETF data and constituent stock data (file: "S&P 500 data (with constituents) for Assignment 2") have been uploaded to Moodle. You are still required to select at least 1,000 daily observations for both the ETF and your chosen stock; however, you no longer need to download the data yourself.

1. Use the provided S&P 500 ETF dataset (tracking the S&P 500 index) from Moodle. Select a time period that includes at least 1,000 daily observations (approximately four years of data, assuming 250 trading days per year). Use the `Adj.Close` prices, which are adjusted for dividends and splits.
2. From the provided dataset, choose one stock whose name starts with the same first letter as your first or last name. This stock must be a constituent of the S&P 500 index and must have at least 1,000 daily observations over the same time period as the ETF selected in (1). Again, use the `Adj.Close` prices.

Assignment tasks

Note: For the tasks below that require hypothesis testing, please use a significance level of 5%.

1. Your first task is to analyse the log-returns of the S&P 500 and of your chosen stock. Log-returns are computed as $R_t = \ln(S_t) - \ln(S_{t-1})$ with S_t denoting the S&P 500 price or selected stock price. Construct log-returns for the S&P 500 and the selected stock. Plot the log-returns of the S&P 500 and a selected stock over the same time period. The x-axis should represent calendar time (years), and the y-axis should display log-returns. Briefly comment on any notable features of the data.

2. Present and compare summary statistics for the S&P 500 ETF returns and chosen stock's returns, including mean, variance, skewness, and kurtosis. Highlight any interesting findings.

3. Create histograms for both, the S&P 500 ETF returns and the selected stock's returns. Overlay each histogram with a fitted normal density with the same mean and variance. Discuss any findings you may find interesting.

4. In many applications, it is assumed for simplicity that stock returns follow a normal distribution. However, your manager doubts this finding and suggests that an alternative distribution with heavier tails than normal might be more appropriate (e.g. Student-t or another suitable distribution). Carefully investigate the manager's conjecture that the return data (both series) follow a Student-t or another distribution rather than a normal distribution. You should use appropriate tests and graphical illustrations to support your conclusions.

5. The manager is also interested in whether annualised returns for the S&P 500 and the selected stock are equal on average. Assist the manager by performing an appropriate test. Comment on the choice of the test and the results.

   Note: An average annualised return can be computed as an average daily return obtained from your sample multiplied by 250, assuming that one year has 250 trading days.

6. Finally, the manager is interested in the magnitude of the annualised mean return on the selected stock and has requested you to test the hypothesis that the annualised average return on the stock is greater than 7%. Comment on the choice of the test and the results.